Data Analytics report

DeepSeek CLI Harness 实测对比

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Codex 与 Pi 在 DeepSeek V4 Flash/Pro、high/max 档位下的 one-shot 性能、缓存效率与实现质量对比。

Technical Summary

同一 prompt、8 个单次样本下，**codex-ds-pro-high** 是默认吞吐首选：10.3 分钟完成且实现可用；**codex-ds-flash-high** 以 94/100 提供最高实现质量，但耗时 21.6 分钟。`max` 不是稳定升级：两组 Pro/max 都超过 31 分钟且均出现关键缺陷。缓存命中率整体很高（96.86%—99.48%），但无法抵消多轮完整上下文重放造成的总输入膨胀。

One-shot 时间：Pro/high 是唯一清晰的速度赢家

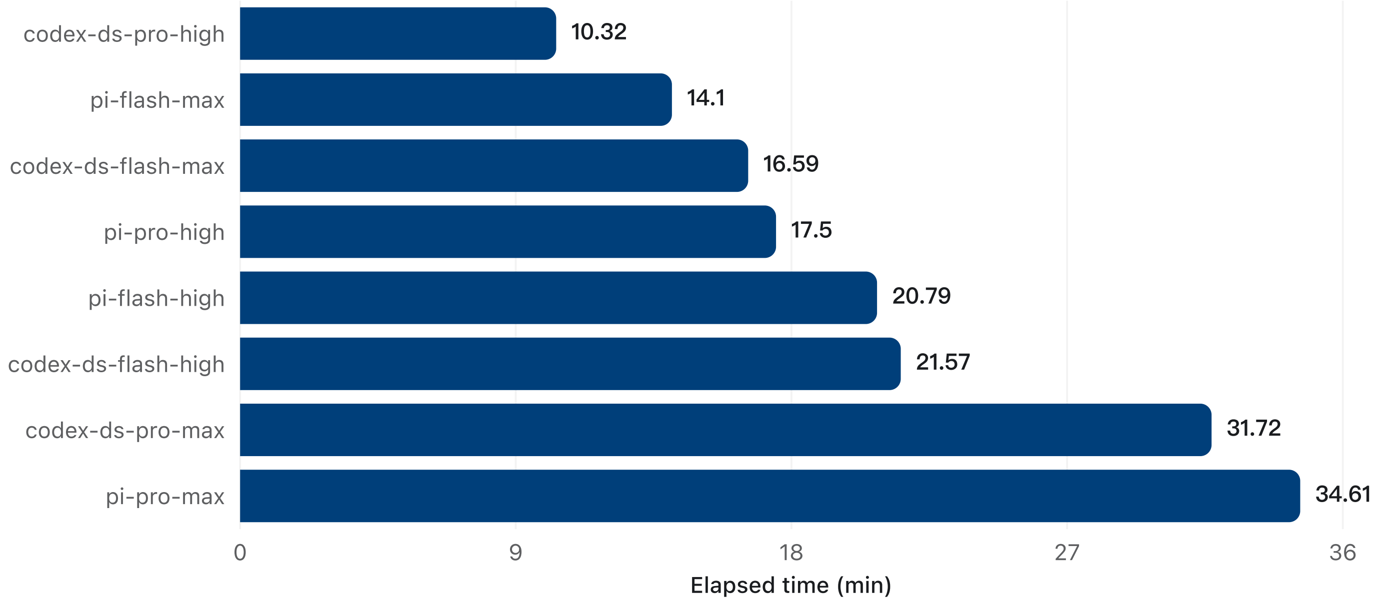
codex-ds-pro-high 用 10:19 完成，比 pi-pro-high 的 17:30 快约 41%。Flash/max 在两个 harness 中都比 Flash/high 快；相反，Pro/max 同时变慢到 31:43—34:37，说明提高 effort 会显著放大循环与验证成本，却没有稳定质量收益。

Source: Normalized benchmark session analysis
Tables: runs/20260813193424/*/.benchmark-runtime/pi/sessions/*.jsonl, runs/20260813195515/*/.benchmark-runtime/{pi,codex}/sessions/**/*.jsonl

One-shot completion time

codex-ds-pro-high finishes in 10.3 minutes; Pro/max stretches past 31 minutes in both harnesses.

Source: Reviewed case-level report query
Tables: analysis/report-source.sql, analysis/analyze-runs.mjs, analysis/quality-assessment.md



缓存：命中率高，但循环体量才是主导变量

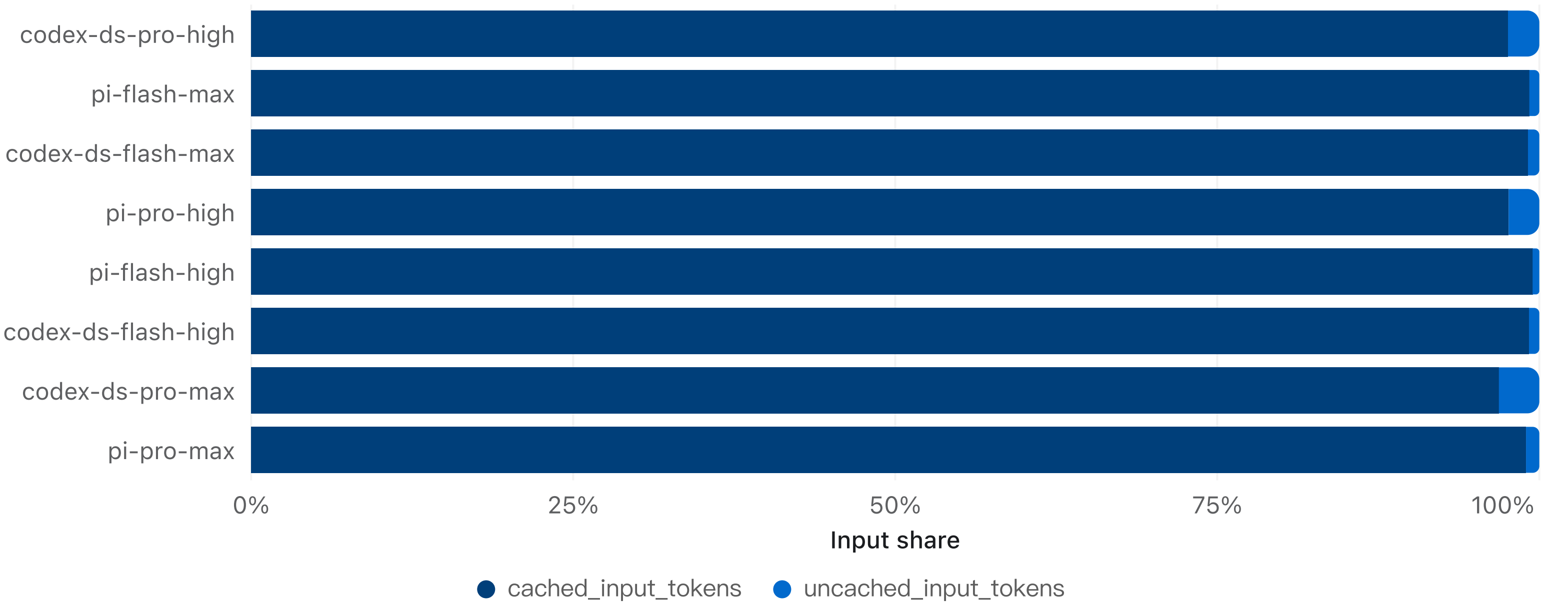
8 个 case 的缓存命中率都很高；真正拉开资源消耗的是重复模型调用后的总输入量。pi-pro-high 只用了 0.67M 总输入和 8 次模型调用，而 pi-flash-high 即使命中率最高，仍累计 3.80M 输入和 41 次调用。高 cache hit 不等于高效率，缓存 token 也不应视为零成本。

Source: Normalized benchmark session analysis
Tables: runs/20260813193424/*/.benchmark-runtime/pi/sessions/*.jsonl, runs/20260813195515/*/.benchmark-runtime/{pi,codex}/sessions/**/*.jsonl

Input token cache composition

Every run exceeds 96.8% cache hit, but total input still ranges from 0.67M to 3.80M tokens.

Source: Reviewed case-level report query
Tables: analysis/report-source.sql, analysis/analyze-runs.mjs, analysis/quality-assessment.md



实现质量：harness 差异小于具体轨迹差异

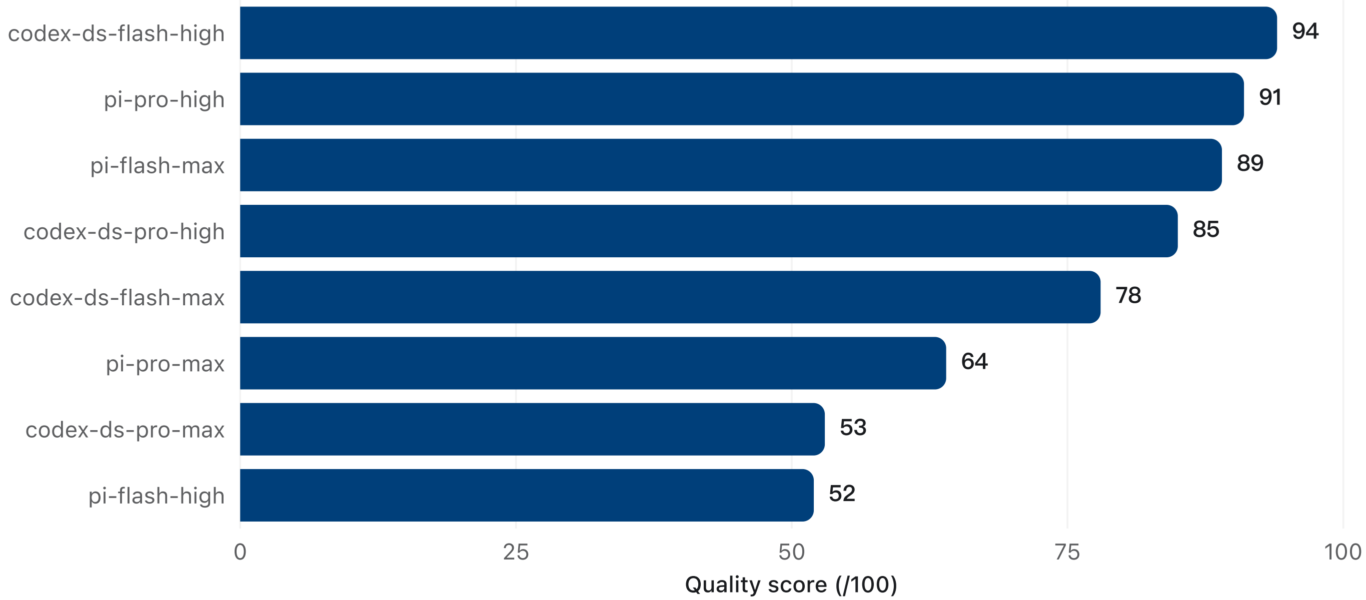
最佳实现是 codex-ds-flash-high（94），最佳视觉构图是 pi-pro-high（91）。pi-flash-high 与 codex-ds-pro-max 都把 ExtrudeGeometry 留在 XY 平面，使软盘竖立；pi-pro-max 在标签淡入路径持续触发 undefined `.style`。这些缺陷跨 harness、跨档位出现，说明结果更受具体执行轨迹与验证策略影响。

Source: Browser and implementation quality assessment
Tables: analysis/quality-assessment.md, runs/20260813193424/*/*.html, runs/20260813195515/*/*.html

Implementation quality score

Five outputs are usable; three contain critical runtime or semantic–geometry defects.

Source: Reviewed case-level report query
Tables: analysis/report-source.sql, analysis/analyze-runs.mjs, analysis/quality-assessment.md



速度—质量前沿：按目标选档，不做单一总榜

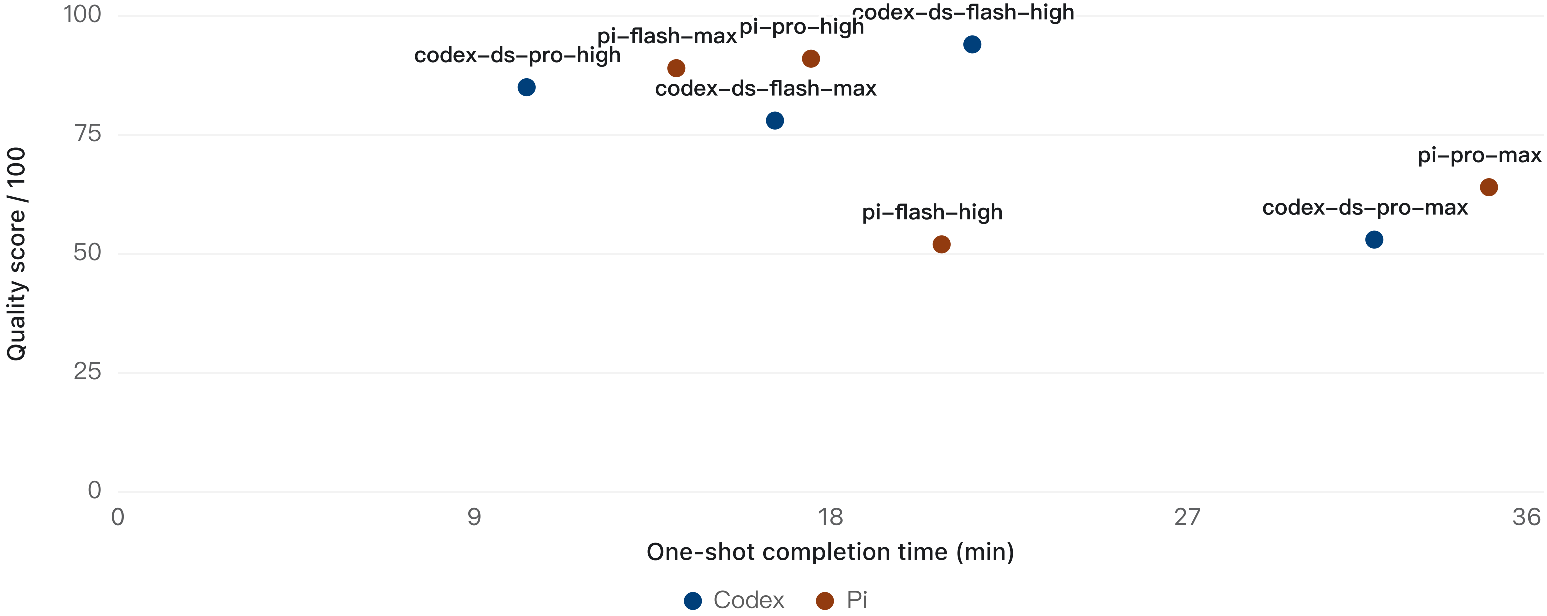
吞吐优先选 codex Pro/high；平衡速度与质量可选 Pi Flash/max；视觉表现优先选 Pi Pro/high；追求最高实现质量选 codex Flash/high。其余 4 个点要么被这些选择支配，要么存在关键缺陷。

Source: Joined performance and quality evidence
Tables: analysis/analyze–runs.mjs, analysis/quality–assessment.md, analysis/chart–map.md

Completion time vs implementation quality

The practical frontier is codex Pro/high for throughput, Pi Flash/max for balance, Pi Pro/high for visuals, and codex Flash/high for maximum quality.

Source: Reviewed case-level report query
Tables: analysis/report-source.sql, analysis/analyze-runs.mjs, analysis/quality-assessment.md



Case 级证据

下表保留完整 token、调用次数、质量分和浏览器发现，便于按 harness、模型档位或 effort 继续复核。

Source: Joined performance and quality evidence
Tables: analysis/analyze-runs.mjs, analysis/quality-assessment.md, analysis/chart-map.md

Case-level metrics and browser findings

Exact session totals; reasoning tokens are a subset of output tokens.

Source: Reviewed case-level report query
Tables: analysis/report-source.sql, analysis/analyze-runs.mjs, analysis/quality-assessment.md

Case	Harness	Tier	Effort	Minutes	Total input	Cached input	Uncached input	Cache hit	Output	Reasoning	Model calls	Tool calls	Quality	Status	Browser finding
codex-ds-flash-high	Codex	Flash	high	21.57	2.57M	2.55M	20.65K	99.2%	32.31K	12.14K	20	25	94	Pass	Best overall implementation; strong exploded view, minor lower-label crowding.
pi-pro-high	Pi	Pro	high	17.5	673.47K	657.41K	16.06K	97.6%	90.88K	79.12K	8	7	91	Pass	Best visual composition; clear layer separation, some right/bottom label crowding.
pi-flash-max	Pi	Flash	max	14.1	2.62M	2.6M	20.39K	99.2%	93.84K	69.12K	29	28	89	Pass	Strong Flash alternative; correct geometry, but exploded framing clips top/bottom labels.
codex-ds-pro-high	Codex	Pro	high	10.32	756.31K	737.92K	18.39K	97.6%	53.08K	32.48K	15	14	85	Pass	Fastest usable result; correct core geometry, denser labels and slight top clipping.
codex-ds-flash-max	Codex	Flash	max	16.59	1.52M	1.51M	13.39K	99.1%	70.66K	46.04K	23	22	78	Pass	Functionally complete; exploded view is over-zoomed with clipped and misleading labels.
pi-pro-max	Pi	Pro	max	34.61	2.91M	2.88M	30.04K	99%	161.56K	133.66K	20	19	64	Critical	Exploded geometry completes, but label animation continuously throws on undefined .style.
codex-ds-pro-max	Codex	Pro	max	31.72	3.53M	3.42M	110.82K	96.9%	135.51K	108.14K	23	32	53	Critical	Semantic geometry failure: extruded shell remains in XY and the floppy stands vertically.
pi-flash-high	Pi	Flash	high	20.79	3.8M	3.78M	19.78K	99.5%	102.33K	79.45K	41	40	52	Critical	Semantic geometry failure: extruded shell remains vertical despite extensive self-tests.

Scope, definitions, and methodology

样本限定为同一 prompt hash 的 8 次 one-shot 完成，每个组合仅 1 次。时间按端到端任务完成口径；Codex 的输入数包含 cached input，因此统一换算为 `uncached = input - cached`。质量采用 30/30/25/15 rubric：功能正确性、几何/规格、视觉构图、验证/可维护性；关键运行时或语义几何缺陷覆盖数值排名。浏览器 QA 使用 1280×720 视口，检查 assembled/exploded、控制台和主要转场；前四名额外完成 exploded→collapsed 往返。

Limitations and robustness

这是每格 n=1 的定向基准，不能估计方差或显著性；结果受瞬时服务负载和 agent 轨迹影响。Pi 会话没有 Codex 同口径的 TTFT，因此未做跨 harness 首 token 比较。Codex 日志未提供可比成本字段，所以报告比较 token 体量而非美元成本。质量分含人工视觉判断，但关键缺陷均由运行时错误或明确的几何轴向证据支持。

Recommended operating policy

1. 默认配置：`Codex + DeepSeek V4 Pro + high`。
2. 视觉质量敏感：优先 `Pi + Pro + high`；若接受更长耗时并追求总体完成度，选 `Codex + Flash + high`。
3. 禁用自动升级到 Pro/max；至少在本任务类型上，它显著更慢且失败率为 100%。
4. 在 harness 回归测试中增加参考视图/轴向断言、CSS2D 标签生命周期测试和 viewport framing 检查；仅靠语法、像素往返或几何数量断言不足。

Further questions

下一轮应对每格至少重复 5 次并记录 p50/p90、失败率、可比 TTFT 与真实计费；再加入 2–3 类任务（纯代码、UI 还原、调试修复）验证当前结论能否跨任务泛化。